

Precalc Lesson 1.11 Homework

Answer
Key

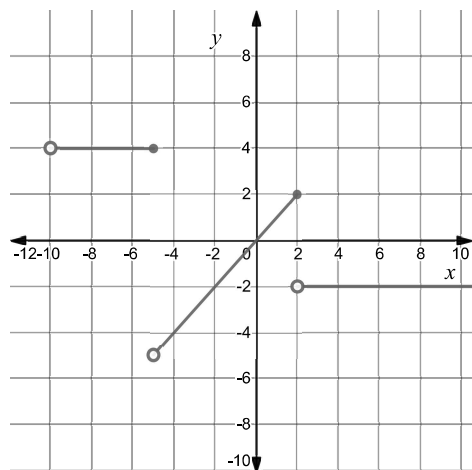
1. $f(-8) = 31$, $f(-4) = -5$, and $f(3) = -8$

2. a. \$11

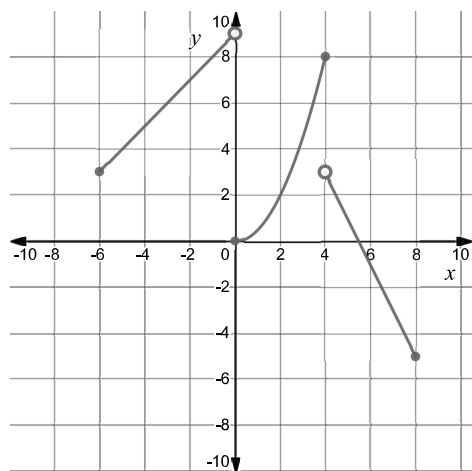
b. \$16.45

c. \$3.49

3.



4.



5. a. $C(p) = \begin{cases} 0.37p & 1 \leq p \leq 74 \\ 0.25p & p \geq 75 \end{cases}$

b. Patty paid $70(0.37) + 10(0.37) = \$29.60$. Had she done them all at once, she would have paid $80(0.25) = \$20$. This mistake cost her \$9.60.

6. $k = 16$

7. C

8. B

9.
$$f(x) = \begin{cases} 6 & -8 < x \leq -4 \\ -\frac{1}{2}x + 2 & -4 < x < 4 \\ 2x - 6 & 4 \leq x \leq 8 \end{cases}$$

10.
$$g(x) = \begin{cases} -x - 3 & -7 \leq x < -3 \\ \sqrt{9 - x^2} & -3 \leq x < 3 \\ -2 & 3 \leq x < 6 \end{cases}$$

Note: In this case the graph would have been the same if $x = -3$ was included in the first branch rather than in the second branch of the function. This is because g is continuous at $x = -3$.